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EEG-based Motor Imagery Classification through Transfer Learning of the CNN

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Abstract— **Brain computer interface (BCI) is a system which is able to translate EEG signals into comprehensive commands for the computers. EEG-based motor imagery (MI) signals are one of the most widely used signals in this topic. In this paper, an efficient algorithm to classify 2-class MI signals based on the convolutional neural network (CNN) through the transfer learning is introduced. To this end, different 3D representations of EEG signals are injected into the CNN. These proposed 3D representations are prepared by combination of some frequency and time-frequency algorithms such as Fourier Transform, CSP, DCT and EMD. Then, CNN will be trained to classify MI-EEG signals. The average accuracy of classification for 5 subjects achieved 98.5% on the BCI competition iii database IVa.**

Keywords— Motor Imagery, EEG Signal, Convolutional Neural Network, Transfer Learning.

Introduction

Brain Computer Interfaces (BCI) are systems by which making connection between brain and machine will be possible. There are three major steps for this goal: preparing brain signals, explicating them, and output to command a machine [1]. Users can control machine activities by BCI without any physical interference and only with thoughts [2]. For instant, it helps disabled people to move their injured muscles, also it is applied in several mind-controlled video games, etc.

EEG signals are described based on frequency bands, amplitude, etc. They are separated into following categories in terms of spectrum: delta 0.5 – 3 Hz, theta 4 – 7 Hz, alpha 8 – 12 Hz, beta 16 – 31 Hz, gamma 32 – 100 Hz, mu 8 – 13 Hz.

These rhythms relate to human physical and mental states. Besides, power of these waves depends on different brain areas from which signals are recorded. Mu and beta rhythms are associated with those cortical areas connected to the brain's normal motor output channels. Therefore, among these waves mu and beta relate to motor imagery signals. These rhythms have the highest power at rest position but whenever a person moves his organs or imagines moving his organs, the power of Mu rhythms starts to change. Hence, motor imagery is one of the most important signals in brain computer interface field. MI signals related to hands, foots, fingers, etc. movements are different from each other. Therefore, we need to separate them. Besides, the nature of these MI- EEG signals are noisy.

In this paper, we tackle to the problem of two tasks of MI signals classification, i.e. right hand/right foot movement separation. Signal preprocessing, feature extraction, and classification are the main steps in this classification problem. Previous studies presented different algorithms for each steps, here we are going to point several proposed methods.

[3] used linear discriminant analysis (LDA) to separate right and left hand MI. It works by providing a linear combination among features. Consequently this method cannot be appropriate for non-linear cases [4, 5]. Support vector machine (SVM) is able to find a non-linear boundary to identify different classes [6, 7]. One of the most prevalent and practical algorithms is common spatial pattern (CSP) to prepare suitable features [7-13]. In most cases, CSP is used with other methods to gain better results. In [7], a combination of CSP and several statistical features were extracted, and then three classifiers such as LDA, CSP and artificial neural network (ANN) were applied to classify right and left hand movement. [14] suggested band pass filter (BPF) and independent component analysis (ICA) to eliminate artifacts. Then, CSP was used as a spatial filter, and finally, classification was performed by applying LDA to separate right and left hand MI. [15] proposed a cross-correlation method to extract features and a least square SVM (LSSVM) to separate two tasks of MI.

Convolutional neural network (CNN) is nearly the most popular way to extract features and classify related datasets [10, 16, 17]. In [16], a hybrid method contains spatial CNN and a long term short memory (LTSM) was suggested. In this study, by using a discrete wavelet transform (DWT) appropriate features were extracted.

In this work, we take an approach for classification of a 2-task MI-EEG signals by a combination of a convolutional neural network, and several time-frequency representations of the EEG signals e.g. CSP, Empirical Mode Decomposition (EMD), discrete cosine transform (DCT), and Fourier transform.

In the following, we firstly introduce the MI dataset, and we explain various domains applied in this study. Then, the proposed classification method is discussed. Finally, experimental results and conclusion are presented in last section.

I. MATERIALS AND METHOD

Here we propose an algorithm based on time-frequency approaches in order to classify two tasks of MI signals. According to Fig.1, different steps of the algorithm will be illustrated. Firstly, we need to prepare EEG Signals in a way that they can be fed into a deep convolutional neural network in order to extract suitable features and also to classify them.

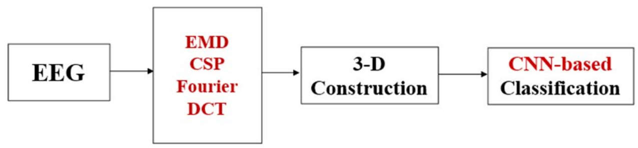

Figure 1. Block diagram of the proposed method based on a CNN feature extractor and classifier of MI EEG-signals.

To this end, some different combination of EEG signals are provided which will be described in detail in next sections.

A. BCI Competition Data Description

In this work, we use EEG signals provided by BCI Competition III by the Berlin BCI group. Dataset Iva of this benchmark is selected and it includes motor imagery signals of right hand and right foot movements [18, 19].They recorded these datasets from 118 electrodes according through the international 10/20 system and by participating of 5 healthy subjects called as aa, al, av, aw, and ay. During this experiment, subjects were supposed to perform the right hand and right foot movement to provide 2-class MI-EEG signals. For each subject, this dataset includes 140 observations per MI task. The experimental setup and information for these dataset is described in [19, 20], in detail.

In Fig.2, the first five observations of class 1 and 2 of subject aa are shown.

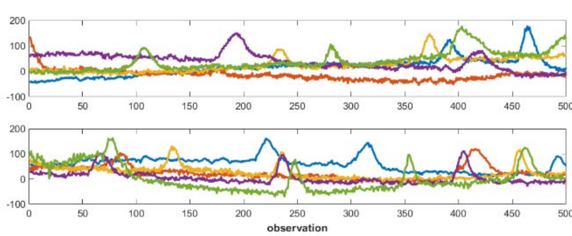

Figure 2. Comparison of five observations of MI signals. The first figure belongs to class 1 and the bottom one belongs to class 2.

B. EMD and Ensemble EMD (EEMD)

One of the most efficient and prevalent approaches is EMD which is considered as a practical way for signal decomposition. In this algorithm, signal is decomposed into plenty of intrinsic mode functions (IMFs) and then these functions are processed by Hilbert Transform, Fourier Transform, Wavelet, etc. Therefore, IMFs will contain useful information of signals and also they play an important role in feature extracting and classification steps in machine learning [21, 22].

Because of noisy and non-preventable movement-based artifacts of EEG signals, it is possible to occur mode mixing phenomenon in EMD [24]. To tackle these problems, Ensemble EMD (EEMD) is proposed [25, 26]. It is based on adding a white noise to the signal and averaging of enough number of trials. Consequently, noise will be removed and this averaged signal will be applied as an IMF component.

In our data processing, the 1st and 2nd IMFs are used to represent the MI-EEG signals in combination with other representations.

C. Common Spatial Pattern (CSP)

CSP is regarded as one of the most effective and common transformations of feature extraction in order to classify different classes of a signal. This transformation is designed in a way that leads to maximizing the ratio of variance between two classes [27], i.e. for example right hand and left hand motor imagery signals.

As a result, CSP-based transformed signals can discriminate two classes of EEG signals [27, 28], appropriately.

D. Data representation

In this work, both feature extraction and classification steps to discriminate two MI-EEG signals are performed by a pretrained CNN, i.e. Alex Net. Therefore, we need to prepare required version of signals to enter the network. We propose an algorithm to construct a 3-dimensional structure out of our EEG signals which will be described here by detail.

We use different transforms of EEG signals such as DCT, CSP, Fourier Transform and IMFs, and calculate their correlations to construct a 2-dimensional frame. After that, these 2-dimensional frames are concatenated to provide 3-D representations to get into AlexNet. Using various transformations, lead to the different structures construction. Different combinations of transforms and consequently different frames constructions are investigated. Finally, the combinations with the best classification results are selected.

Given an original 1-dimensional EEG signals named *s*, several representations of it including DCT(s), CSP(s), IMF1(s), IMF2(s), IMF3(s), F(s), and Ph(s) are introduced in Table 1.

Table1. Introducing different representation of EEG signal

s	Original signal
* DCT(*s*)	Coefficients of Discrete Cosine Transform of s
CSP(*s*)	Magnitude of CSP-based representation of s
IMFk(*s*)	kth IMF of s based on EEMD
F(*s*)	Product of Ph(s) and the magnitude of Fourier transform of s
Ph(*s*)	Phase of Fourier transform of s

*The DC component of DCT(s) has been eliminated and the rest of it has been cropped in suitable windows.

We need to combine these 1-dimensional signals such as *x* and *y* together to provide 2-dimensional structures

$f(x,y)=xy^T$. Now, we just need to combine three of these 2-D frames to construct a 3-D structure.

II. FEATURE EXTRACTION AND CLASSIFICATION

The most important step in each classification process, is selecting and then extracting several suitable and discriminative features of signal. In this study, required features will be given by a pre-trained CNN named Alexnet which is chosen among different pre-trained CNN like LeNet, VGG, Google Net, etc.

AlexNet was trained on beyond a million images from the ImageNet database [29], and it can separate them into 1000 various categories. Therefore, it has strong ability for feature representation. It includes a cascade of five convolutional layers and three fully connected layers [30]. As it can be seen in Fig. 3, there are some max-pooling layers after some of the convolutional layers. AlexNet is designed in a way that it needs an input image size of 227-by-227. So, it turns out that we need to form the MI-EEG signals into 3-dimensional structures in size of 227-227. Besides, the size of employed filters in different layers of Alex net is shown in Fig. 3.

In this work, the filters of convolutional layers and also weights of FC layers are initialized from its original version. Then, learning has been applied to MI-EEG signals classification through the proposed 3-D structures.

III. EXPERIMENTS AND RESULTS

As discussed in section 2, we applied BCI Competition III dataset Iva [19], a two-class MI EEG signal, to implement our proposed method. It contains right hand and right foot motor imagery of 5 subjects, and 280 observations for each subject.

We allocated randomly 70% of observations/subject (i.e. 196 observation) to training set, and the rest of them for evaluating the proposed algorithm.

To apply AlexNet as a classifier, we should replace three last layers of the network with a Fully-Connected layer, Softmax layer and a classification layer, respectively. Besides, we need to set the number of classes in softmax, by default it was arranged to tackle a 1000-class problem, but here we have a 2-class discrimination. Therefore, it is necessary to set the component to a 2-class discrimination.

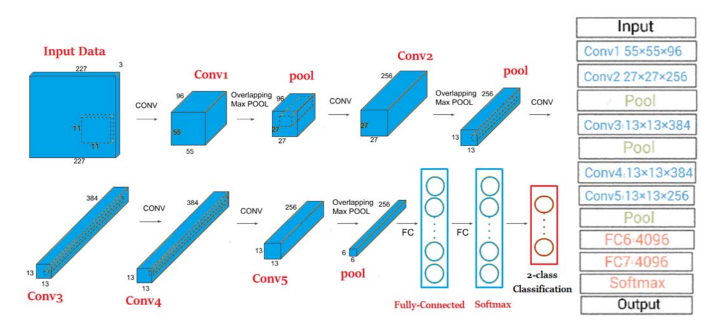

Figure 3. AlexNet: left-architecture and right-different layers

One of the most important criteria for determining the efficiency of the classification algorithm is the accuracy or classification rate (Classification Accuracy) by which the overall accuracy of the classifier is calculated. Classification accuracy is obtained by using equation 1, where TP, TN, FP and FN are referring to true positive, true negative, false positive and false negative respectively.

The sum of TP and TN should be maximized here in our 2-class problem.

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

The average Accuracy of multiple runs will be announced as a final single result.

Several experiments are designed to evaluate the proposed method for classification of two-class MI EEG signals.

Experiment I: For our first experiment, we examine the various domains from which mentioned 3-D structures provided out of only one component. In table 2, the average results of classification of MI-EEG signal based on one domain are shown.

Table2. The average classification accuracy (%) in self-combination mode.

	Raw Signal	DCT-based Coefficients	CSP-based Coefficients	F(s)	Ph(s)	M1*	M2
Test Set	50	47.6	73.2	97.6	51.1	57.1	47.6

*Mk refers to IMFk

In case that the 3-D input required for the network is produced out of only one domain, the Fourier domain-based representation performs better than others, although there are helpful and discriminative information in all of these representations. Therefore, in following experiments we use combination of different domain to provide various input, and we examine them to gain the best results.

Fig.4 illustrates an example of AlexNet training process by applying a 3-D input made out of only Fourier Transform component, i.e. F(s).

Experiment II: In this experiment, different combinations of introduced representations are applied to construct various triple-frames by putting three of representations together. In table 3, these combinations are presented.

Table.3 Different combinations of various EEG representation

$Z_1 = f(s,s)$	$Z_7 = f(s,c)$	$Z_{13} = f(d,c)$	$Z_{19} = f(c,ph)$
$Z_2 = f(d,d)$	$Z_8 = f(s,M_1)$	$Z_{14} = f(d,M_1)$	$Z_{20} = f(c,M_1)$
$Z_3 = f(c,c)$	$Z_9 = f(s,M_2)$	$Z_{15} = f(d,M_2)$	$Z_{21} = f(c,M_2)$
$Z_4 = f(f,f)$	$Z_{10} = f(s,ph)$	$Z_{16} = f(d,F)$	$Z_{22} = f(F,M_1)$
$Z_5 = f(M_1,M_1)$	$Z_{11} = f(s,F)$	$Z_{17} = f(d,ph)$	$Z_{23} = f(F,M_2)$
$Z_6 = f(M_2,M_2)$	$Z_{12} = f(d,s)$	$Z_{18} = f(c,F)$	$Z_{24} = f(M_1,M_2)$

Now, we examine our proposed method by feeding network these triple frames ($Z_kZ_lZ_m$) to extract features and classify. Same as first experiment, the result shown in table.4 are from subject 1 (aa) across multiple runs of the algorithm.

Table 4. The average classification accuracy (%) of subject aa by selected domain

	$Z_{14}Z_{15}Z_{16}$	$Z_{01}Z_{10}Z_{04}$	$Z_{02}Z_{16}Z_{01}$	$Z_{16}Z_{16}Z_{16}$	$Z_{95}Z_{24}Z_{06}$	$Z_{24}Z_{10}Z_{14}$	$Z_{17}Z_{11}Z_{16}$
Test Set	94	100	96.4	100	42.6	52.3	100

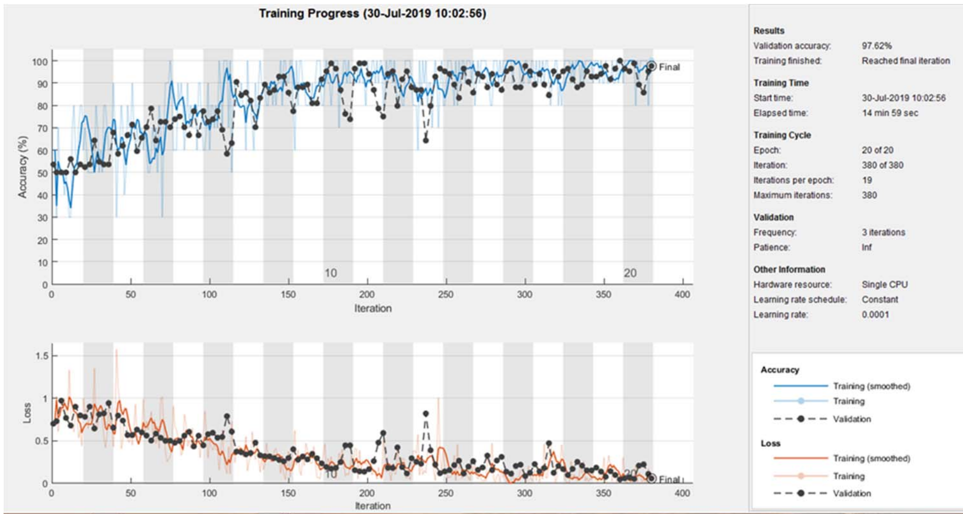

Figure 4. The training process in AlexNet in Fourier domain features, F(s)

According to table 4, the results of different hybrid structures, as AlexNet inputs, to classify MI signals by network in three modes $Z_{01}Z_{10}Z_{04}$, $Z_{16}Z_{16}Z_{16}$ and $Z_{17}Z_{11}Z_{16}$ have yielded the best classification accuracy of 100% for subject aa and channel F4 of EEG. As an example Fig. 5 shows the process of training of MI signals of subject aa by AlexNet with $Z_{14}Z_{15}Z_{16}$ as input. Channel F4 of EEG is selected and after 20 times of training, it has reached the accuracy of 94%.

Experiment III: Based on the previous experiment, the 3-D structures as CNN inputs with the best results were selected and findings for all 5 subjects are listed in table.5.

According to table 5, the $Z_{11}Z_{16}Z_{17}$ is one of the best 3-D structures to enter AlexNet.

Here, in table.6 we compare our proposed method with other works in which average classification rates of dataset Iva of BCI Competition III are shown.

Table 5. The results of classification of all 5 subject by proposed method (%)

	$Z_{14}Z_{15}Z_{16}$	$Z_{16}Z_{16}Z_{16}$	$Z_{11}Z_{16}Z_{17}$
Subject aa	94	100	100
Subject al	100	98.8	97.6
Subject av	70.2	89.3	98.8
Subject aw	89.3	97.6	96.4
Subject ay	70.2	89.3	100
Avg.	84.75	95	98.57

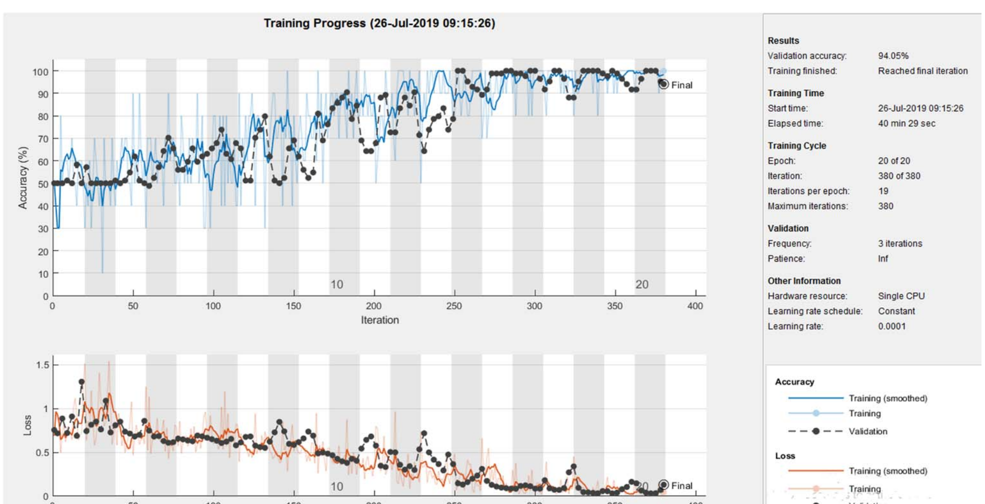

Figure 5. The training process of subject aa in AlexNet with $Z_{14}Z_{15}Z_{16}$ as input

Table 6. Comparison of gained classification accuracy (%) with other methods

	[31]	[32]	[33]	[12]	[11]	[13]	The Proposed Methods
Subject 1	97.92	97.78	96	68.10	66.79	86.61	100
Subject 2	97.88	98.89	92.3	93.88	96.07	100	97.6
Subject 3	98.26	96.67	88.9	68.47	52.14	66.84	98.8
Subject 4	94.47	98.89	95.4	90.58	71.43	90.63	96.4
Subject 5	93.26	95.56	91.4	84.65	50	80.95	100
Average	**96.36**	**97.56**	**92.8**	**81.14**	**67.29**	**85.01**	**98.57**

IV. CONCLUSION

In this paper, a method was proposed to classify two-class MI-EEG signals and separate right hand and right foot motor imagery tasks. We used BCI Competition III dataset Iva which contains data of 5 subjects and it was recorded from 118 EEG channels. A pre-trained convolutional neural network was trained to classify considered signals through transfer learning. The challenge is to transform input 1-D signals into 3-D image to be fed to the selected network. Therefore, we applied different representations of signal in various domain, like DCT, EEMD, CSP and Fourier Transform. The findings were strongly impressive. By designing several experiments, we could evaluate our proposed algorithm. The method gained classification accuracy of 100% for separating two MI tasks of subject aa and average accuracy of 98.5% of all 5 subjects which is a great result in compare to other methods.

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